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Ethics and Bias in Natural Language Processing

① Ethical Risks in NLP
② Algorithmic and social bias
③ Harm and misclassification
④ Bias auditing and mitigation
⑤ Governance and responsible deployment
⑥ Summary and Takeaways

Objectives

- Identify core **ethical risks in NLP**: bias, privacy, misuse, dual use.
- Understand sources of **algorithmic and social bias** in data and models.
- Analyze real-world case studies of **harm and misclassification**.
- Examine methods for **bias auditing and mitigation**.
- Evaluate governance and responsibility in deploying LLMs.

Ethical Risks in NLP

Core Ethical Challenges in NLP Systems

- **Algorithmic:** Occurs when NLP models reproduce or amplify biases present in training data.
  - Sources: **Representation bias** (imbalanced datasets), **historical bias** (social inequalities embedded in data), **measurement bias** (incorrect task formulation).
- **Privacy and Surveillance:** NLP systems can inadvertently **memorize and leak private data** (e.g., names, addresses).
  - Issue in models like GPT-2/3 that memorize rare training sequences.
- **Misinformation and Abuse:** NLP powers deepfakes, synthetic news, and propaganda bots.
  - Example: LLMs generating persuasive fake news articles.
- **Transparency and Accountability:** Models are black-box; **difficult to audit** or explain decisions.
  - Challenge in high-stakes settings: healthcare, law, hiring.
- **Societal Impact:** Deployed NLP systems affect job prospects, court decisions, mental health tools—raising **questions of fairness and liability**.

Algorithmic and social bias

❌ Gender Bias in Resume Screening

- **System Description:** Amazon trained a model (2014–2017) to identify top candidates from resumes.
- **Bias Manifestation:** Penalized resumes with terms like "women's chess club captain"; favored male-coded language.
- **Technical Root Cause:**
  - Training data skewed by historical male-dominated tech hiring.
  - Model implicitly learned **gender-correlated proxies**.
- **Outcome:** Project scrapped before deployment after internal audits.
- **Lesson Learned:**
  - **Bias amplification** happens silently unless proactively checked.
  - Even unsupervised language features (e.g., embeddings) can encode gender/race cues.
- Reference: Reuters investigation, 2018

Stereotypes in Generative Language Models

- **Problem Observed:**
  - Prompting GPT-3 with phrases like "Two Muslims walked into…" often led to **violent or extremist completions**.
  - Women, minority groups stereotyped by profession or attributes.
- **Technical Cause:**
  - Pretraining on large-scale internet text, where **unmoderated data (e.g., Reddit, Common Crawl)** embeds cultural biases.
- **Reinforcement through Scaling:**
  - Larger models are **more fluent but not necessarily more fair**.
- **Ethical Challenge:**
  - Bias is hard to detect during generation (no static labels).
  - Debiasing post-hoc is non-trivial; can affect model fluency and alignment.

References: Bender et al., 2021: *On the Dangers of Stochastic Parrots*

❌ Real-World Harm from Faulty Translation

- **Incident:** Arabic phrase by a Palestinian worker auto-translated by Facebook to “attack them” instead of “good morning.”
- **Consequence:** Arrest and detainment by Israeli police; released after verification.
- **Technical Problem:**
  - Lack of dialectal distinction (Modern Standard Arabic vs. colloquial).
  - Absence of **context-aware translation** for informal language.
- **Wider Issue:**
  - Overreliance on NLP in **high-risk domains (security, law)** without human verification.
- **Postmortem:** Facebook apologized and improved moderation filters.

Ethical Takeaway:

- NLP systems can create **false evidence** in legal contexts.
- **Accountability gaps** when AI outputs are taken at face value.
- Reference: The Guardian, Oct 2017

❌ Racial Bias in Sentiment Scores

- **Problem:** Sentiment analysis API produced **more negative scores** for sentences with African-American names or dialect features (e.g., AAVE).
- **Root Cause:**
  - Training data lacked **racial linguistic diversity**.
  - Models treated non-standard English as informal or negative.
- **Technical Insight:**
  - Word embeddings (e.g., Word2Vec, GloVe) reflect social biases through **co-occurrence statistics**.
  - Example: "Black-sounding" names closer in vector space to negative words.
- **Implications:**
  - Deployed APIs can have **discriminatory downstream effects** in hiring, policing, and credit scoring.
- **Research Response:**
  - Techniques like **debiasing embeddings**, counterfactual data augmentation, and dialect-aware modeling.

References:

- Sap et al., 2019: *The Risk of Racial Bias in Hate Speech Detection*

Dual Use: What is Dual Use?

- **Dual-use technology**: Developed for beneficial purposes but can also be exploited for harmful ends.
- Common in fields like chemistry, biology, and now AI/NLP.
- NLP dual use: Text generation, translation, summarization, dialogue systems, etc.

Examples:

- Chatbots for mental health vs. chatbots for manipulation/scams
- Summarization tools for accessibility vs. condensing propaganda
- Machine translation for communication vs. spreading disinformation

Large Language Models (LLMs) as Dual-Use Systems

- **Positive Uses:**
  - Code generation, educational tutoring, writing assistance, accessibility support.
- **Negative Uses:**
  - Generating phishing emails, malware code, spam, and disinformation at scale.
- **Key Insight:**
  - LLMs lower the **barrier to entry for sophisticated attacks**, previously requiring expertise.

Real-World Example:

- GPT-2 withheld at first release due to fears of misuse in fake news generation.

Harm and misclassification

Known Misuses of NLP Technologies

Documented and Predicted Threats

- **Automated Disinformation Campaigns**
  - Synthetic political content, fake reviews, astroturfing.
- **Scalable Social Engineering**
  - Auto-generated scam/phishing emails with contextual relevance.
- **Toxicity and Hate Speech Amplification**
  - Repetition and refinement of harmful language and ideology.
- **Deepfake Text for Legal or Academic Fraud**
  - Forged articles, legal arguments, or evidence in digital form.
- **Reference:** Brundage et al., 2018 – *The Malicious Use of AI*

Phishing via AI-generated Emails

Demonstration of Dual-Use Potential

- GPT-3 used to generate **highly personalized spear-phishing emails**:
  - Style, grammar, and context far more convincing than template-based attacks.
- **Experiment Findings:**
  - Participants found AI-generated emails harder to distinguish from real ones.
- **Implication:**
  - NLP-driven attacks may **outpace current defenses**.
- **Reference:** Kumar et al., 2021 – *Fine-Tuned Language Models Are Zero-Shot Phishers*

The Cost of Misclassification

- **Misclassification is not just a technical error — it can have real-world consequences.**
- **High-Stakes Domains:**
  - **Healthcare:** Mislabeling a condition in clinical notes or diagnostic NLP tools can delay treatment or cause harm.
  - **Hiring and Admissions:** Resume parsers may misclassify candidates due to race- or gender-coded language, reinforcing systemic inequality.
  - **Content Moderation:** Classifying dialects like AAVE (African American Vernacular English) as toxic speech leads to disproportionate silencing.
  - **Legal and Surveillance Applications:** NLP tools used in crime detection or risk assessment may misclassify harmless language as threatening, especially for marginalized groups.

Understanding Social Context

- **Language is socially situated**
  - The same phrase may mean different things depending on who says it, where, and in what context.
- **Cultural and Linguistic Variation**
  - Models trained on standard or majority-language corpora often fail on non-dominant dialects and multilingual settings.
- **Historical and Power Dynamics**
  - Misclassification tends to disproportionately affect those already at social disadvantage — compounding harm.
- **Key Insight:**
  - Technically equivalent errors can have **radically different impacts** depending on **who is misclassified** and **what the consequences are**.

Understanding Social Context

Learning to Assess AI Systems Adversarially

- **Why Adversarial Assessment Matters:**
  - AI systems don't fail uniformly — they often fail **unfairly**
  - Critical analysis helps uncover **hidden assumptions, vulnerabilities, and harms**
  - Especially important for **NLP**, where language is inherently social and sensitive
- **Key Assessment Dimensions:**
  - **Ethics of the Research Question**
    - Is the task itself ethically sound?
    - Does it risk harm by design (e.g., inferring private traits like political beliefs or sexual orientation)?
  - **Impact & Dual-Use Risk**
    - Who benefits from this technology?
    - Who could be harmed — intentionally or unintentionally?
    - Could releasing this model/data reshape lives, policies, or institutions?
    - Are safeguards in place to prevent misuse?
  - **Privacy and Consent**
    - Who owns the data? Was it collected ethically?
    - Public ≠ publicized: just because it's online doesn't mean it's free to use
    - Have user expectations been respected?

Learning to Assess AI Systems Adversarially

- **Key Assessment Dimensions, cont.:**
  - **Bias in Data**
    - Are there hidden artifacts that shortcut learning (e.g., dataset leaks, label leakage)?
    - Is the population representative (age, gender, dialect, region)?
    - How was the data annotated, and by whom?
  - **Bias in Models**
    - Are there confounding variables affecting outputs (e.g., name = race)?
    - Is the system optimizing the **right** objective (fairness vs. raw accuracy)?
    - Does it perform robustly on **corner cases**, minority examples, and ambiguous inputs?
  - **Beyond Accuracy: A Utility-Based View**
    - Accuracy hides **false positive / false negative asymmetries**
    - Misclassifying hate speech as benign ≠ misclassifying a joke as offensive
    - Consider fault tolerance: is the system resilient to unexpected inputs or shifts?

Algorithmic: Social Bias in Data and Models

- **What is Algorithmic Bias?**
  - Systematic error in AI outputs due to **social biases embedded in data or models**
  - Emerges when models reflect, reproduce, or **amplify societal stereotypes**
  - Especially insidious in **language models**, where word choice encodes social meaning
- **Social Bias in Language Data:**
  - Everyday language reflects social categories (gender, age, class)
  - Models learn these patterns **without understanding social context or fairness**
- **Examples:**
  - Gendered language:
    - "giggle" vs. "laugh" → more likely attributed to women
    - "brutal" vs. "fierce" → stereotypically gendered usage
  - Age-related framing:
    - "amazing" vs. "impressive" → differentially used with younger vs. older individuals
  - Class associations:
    - "suggestions" vs. "proposals" → linked to perceived occupational status

Algorithmic: Social Bias in Data and Models

- **Why Does This Happen?**
  - **Language is not neutral**
    - Training corpora reflect media, social media, literature — all shaped by culture and power
  - **Co-occurrence ≠ causality**
    - If "nurse" co-occurs with "she," the model may assume all nurses are women
  - **Default social groups**
    - We intuitively recognize a "default": male, young, able-bodied, white, educated
    - These groups become **normative baselines** in model reasoning
- **Implicit Bias in NLP Models:**
  - Biases in data become **latent representations** in embeddings and predictions
  - Often undetectable until surfaced through probing or downstream failures
- **Consequences:**
  - Harmful stereotypes in generation (e.g., toxic completions for minority prompts)
  - Unequal performance across demographic groups
  - Reinforcement of structural inequities in real-world applications

Psychological Perspective on Cognitive Bias

- **Why Bias Is Inevitable — Even Before AI:** Cognitive bias originates from the **human mind's built-in mechanisms** for navigating a complex world.
- **Cognitive Mechanisms Behind Bias Formation:**
  - **Categorization as a Shortcut**
    - The mind simplifies reality by grouping stimuli into categories
    - Enables fast decision-making, but can flatten individual variation
  - **Schemas and Mental Representations**
    - Prior experiences are abstracted and stored as **schemas**
    - These guide interpretation, memory, and behavior
    - Once formed, schemas are **resistant to change**
  - **Automatic Activation**
    - Encountering a category (e.g., gender, profession) unconsciously triggers associated schema
    - Can lead to **automatic and unexamined judgments**, even without intent
- **Cognitive bias** is a systematic pattern of deviation from rationality in judgment, often driven by schema-based thinking.
- **Implications for NLP:**
  - AI models trained on human data **inherit these cognitive shortcuts**
  - Word embeddings, language models, and classifiers **replicate associative patterns** learned from biased corpora
  - NLP systems may reflect or even amplify stereotypes **not because they are designed to**, but because they are **trained to mimic human language**

Biases That Affect Human Decision-Making

- Understanding how humans systematically err in judgment helps us recognize how these biases may be **mirrored or magnified in AI systems**, especially in NLP tasks involving classification, recommendation, or evaluation.
- **Key Cognitive Biases:**
  - **Confirmation Bias**
    - Tendency to seek, interpret, and recall information that supports preexisting beliefs
    - *Impact in NLP:* Models may reinforce user biases in search, summarization, or recommendation systems
  - **Ingroup Favoritism**
    - Preference for individuals perceived as belonging to one's own social group
    - *Impact in NLP:* Sentiment or trust models may favor dialects or cultural norms of majority groups
  - **Group Attribution Error**
    - Assuming that behaviors of a few group members reflect traits of the entire group
    - *Impact in NLP:* Stereotyping in generated text, mislabeling minority-group language as toxic or threatening

Biases That Affect Human Decision-Making

- **Halo Effect**
  - The tendency for positive impressions of a person or concept to influence judgments of unrelated traits
  - *Impact in NLP:* Positive sentiment toward authoritative sources may mask misinformation
- **Just-World Hypothesis**
  - Belief that people get what they deserve; often leads to victim-blaming
- *Impact in NLP:* Models trained on biased discourse may encode harmful social narratives

Other Notable Biases:

- **Availability Heuristic:** Overestimating the importance of information that comes readily to mind
- **Anchoring Bias:** Overreliance on the first piece of information encountered
- **Fundamental Attribution Error:** Over-attributing others' actions to character rather than context

Social Stereotypes in NLP Systems

- Simplified, generalized beliefs about members of a social group
- Often **culturally ingrained**, **implicitly held**, and **resistant to change**
- Can manifest in **lexical associations**, **syntactic framing**, and **semantic connotations**
- **Common Axes of Stereotyping in NLP:**
  - **Gender:** E.g., associating "nurse" with women and "engineer" with men
  - **Race:** E.g., African American Vernacular English (AAVE) misclassified as offensive language
  - **Disability:** E.g., metaphoric use of disability terms (e.g., "crazy", "lame") in negative contexts
  - **Age:** E.g., assuming older people are technologically illiterate or resistant to change
  - **Sexual Orientation:** E.g., NLP models trained on biased forums may associate LGBTQ+ identities with derogatory terms
  - **Culture:** E.g., Western-centric framing of politeness, formality, or emotion
  - **Class / Socioeconomic Status:** E.g., framing poverty as a moral failure or lower intelligence
  - **Poverty:** E.g., associating lower-income communities with crime or dysfunction in news summaries
  - **Language / Dialect:** E.g., standard English privileged over regional or ethnic varieties
  - **Religion:** E.g., Islamophobic associations in LLM completions (e.g., “Muslim” → “terrorist”)
  - **National Origin:** E.g., framing certain nationalities as threats or using dehumanizing metaphors

Bias in Data and Language for NLP

- **Bias in Datasets: From Collection to Annotation**
  - **Data Selection / Sampling Bias**
    - Training corpora overrepresent certain demographics, dialects, topics, or platforms (e.g., Reddit, Wikipedia)
    - Leads to **unequal model performance** and **underrepresentation of minority groups**
  - **Annotator Selection Bias**
    - Annotation often done by crowdworkers from specific geographic, cultural, or educational backgrounds
    - Their worldviews shape how they label ambiguity, toxicity, sentiment, etc.
  - **Annotators' Cognitive Biases**
    - Personal schemas and implicit associations influence labeling
    - E.g., labeling passionate speech by Black users as "angry" or "aggressive" more often than equivalent speech by white users

Bias in Data and Language for NLP

- **Bias in Language: Where Social Norms Meet Text**
  - **Stereotypes and Prejudices**
    - Texts often encode gender, racial, cultural, and other social biases (e.g., “bossy” for women, “terrorist” for Muslims)
  - **Toxic Language and Hostility**
    - Hate speech, slurs, and discriminatory framing are disproportionately targeted at marginalized groups
  - **Historical Human Biases**
    - Language corpora reflect **structural inequalities**, outdated norms, and power imbalances over time
  - **Human Reporting Bias**
    - People write about what’s *interesting*, *shocking*, or *politically charged*, not what's representative
    - Word frequencies and topic salience do **not** mirror the real world

three black teenagers

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https://www.bbc.com › world-us-canada-36487495

'Three black teenagers' Google search sparks Twitter row

Jun 9, 2016 — A **Google image search for "Three white teenagers"** throws up photos of happy, smiling groups of friends.

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Jun 9, 2016 — **Google image searches for "three black teenagers"** and "three white teenagers" get very different results, raising troubling questions about how racial bias in ...

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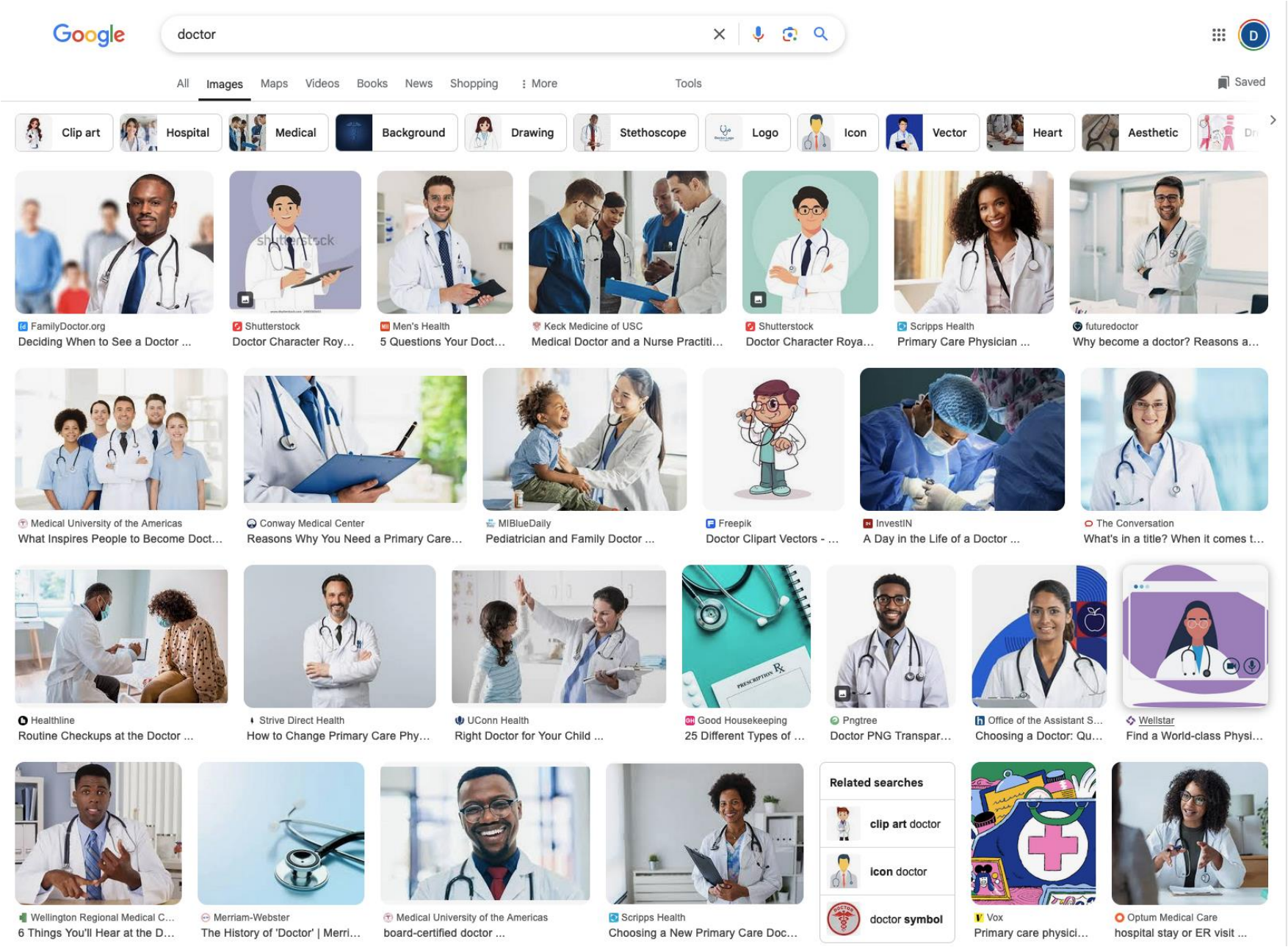
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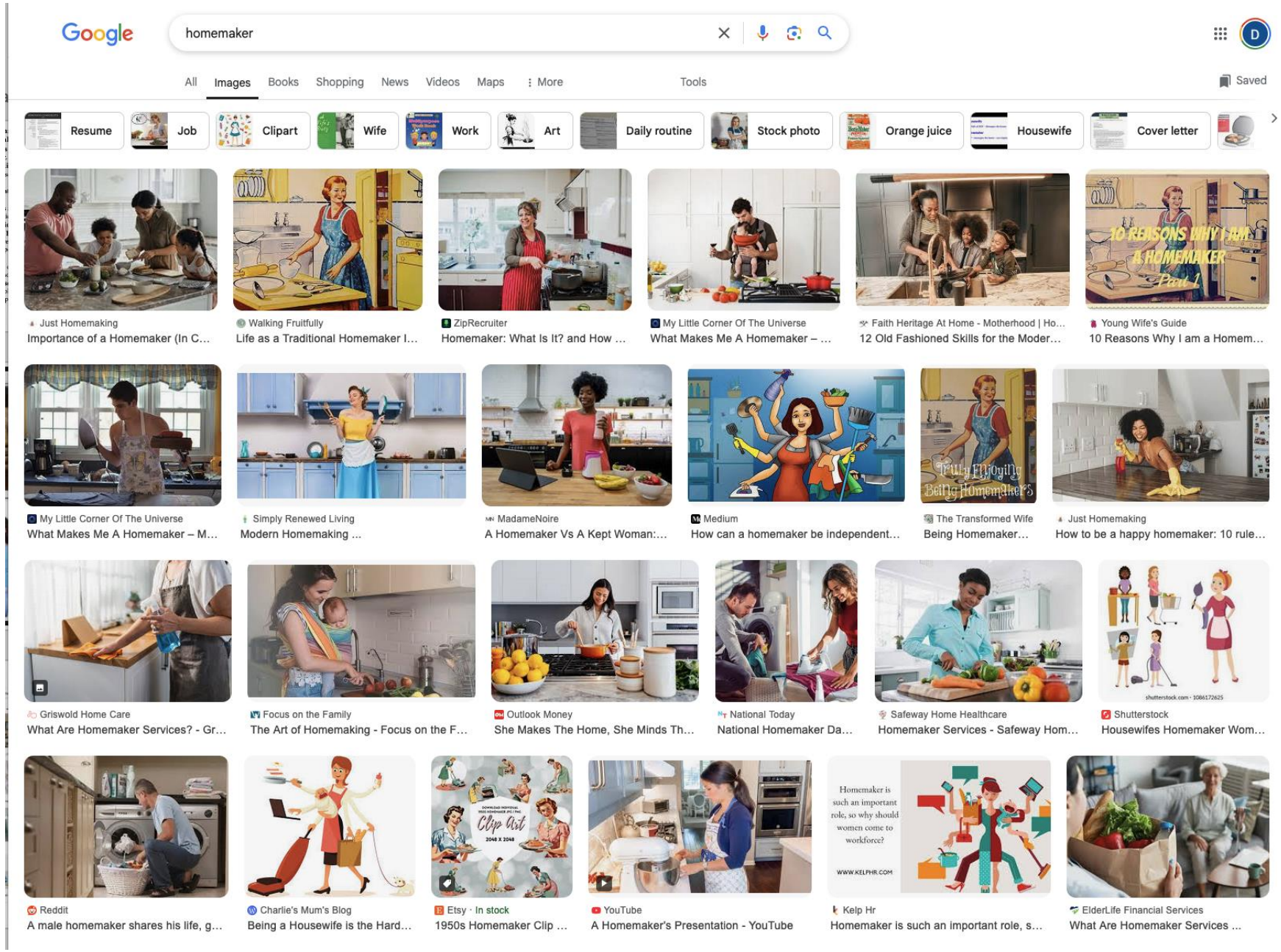
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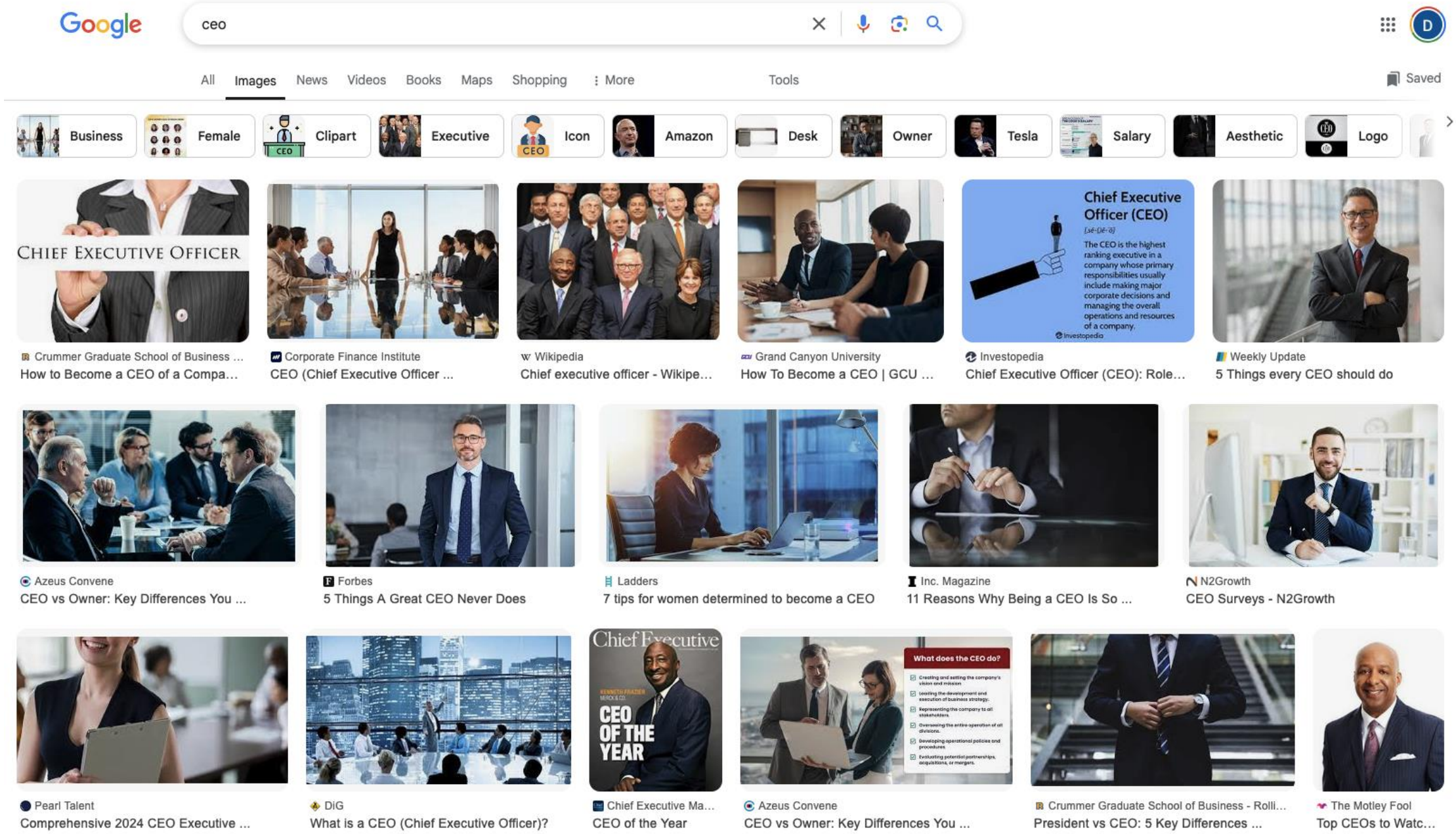
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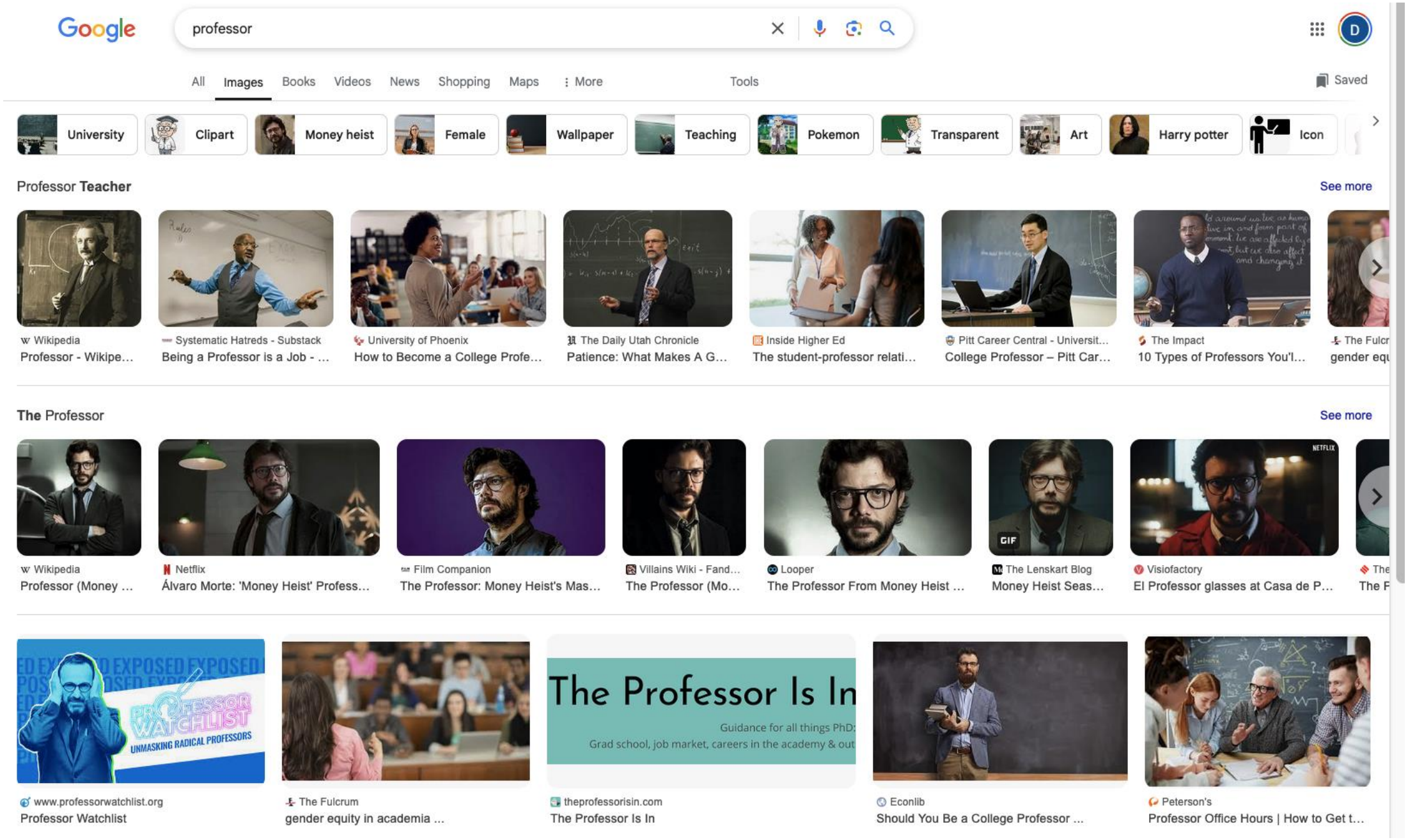
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Case Study – "AI Gaydar"

Overview of the Study:

- 2017 study by Wang and Kosinski claimed that a deep learning model could predict sexual orientation from facial images with over 80% accuracy.
- Labeled “AI Gaydar” by media and critics.

Key Claims:

- The model identified “facial morphology” differences correlated with sexual orientation.
- Claimed higher accuracy than human judges.

Technical Methods:

- Input: Profile photos from dating websites labeled by users’ stated orientation.
- Architecture: CNN trained on visual features.
- Not NLP per se, but shares **ethical vulnerabilities** relevant to language modeling:
  - Inferring protected attributes.
  - Misuse of large-scale web data.
  - Lack of consent and context.
  - Risk of reinforcing stereotypes.

Ethical and Methodological Critiques

- **Major Criticisms:**
  - **Spurious correlations:** Backgrounds, image styles, or grooming habits may be mistaken for inherent traits.
  - **Privacy violation:** Photos scraped without informed consent.
  - **Reinforces dangerous pseudoscience:** Echoes historical abuses in phrenology and physiognomy.
  - **Deployment risks:** Authoritarian use for surveillance or persecution of LGBTQ+ individuals.
- **Expert Responses:**
  - GLAAD, HRC, and numerous ethicists condemned the study.
  - Argued it could embolden discrimination and policy abuses.
- **Key Ethical Lapses:**
  - Ignoring downstream harm.
  - No stakeholder consultation.
  - Lack of robust causal inference or fairness auditing.

Data & Privacy Concerns in the "AI Gaydar"

- **Photos downloaded from a popular American dating website**
  - Researchers scraped publicly accessible profile images from dating sites.
  - No direct user consent or opt-in for research purposes.
- **Legal ≠ Ethical**
  - The data collection may have been technically legal under U.S. copyright/data laws.
  - However, legality doesn't imply moral acceptability or compliance with ethical norms.
- **Public ≠ Publicized**
  - Being online doesn't mean people expect their data to be used for research.
  - Expectation of privacy remains even in semi-public digital spaces.
- **Did these individuals agree to participate in the study?**
  - No informed consent process, no participant protections, no opt-out mechanism.
  - No opportunity to review or contest model usage or findings.

  → **Violation of the Social Contract**
  - Undermines public trust in researchers and AI systems.
  - Exploits marginalized communities under the guise of scientific inquiry.
  - Violates principles of **respect for persons** and **beneficence** in research ethics.

Lessons for NLP and Beyond

- **Shared Concerns with NLP Models:**
  - Inference of private attributes (e.g., gender, religion, political stance) from text embeddings or language style.
  - Risks of misuse in surveillance, profiling, or discrimination.
  - Lack of model interpretability compounds harm.
- **Takeaways:**
  - Just because something *can* be predicted doesn't mean it *should*.
  - Ethical NLP requires understanding **intent, consent, and consequence**.
  - Importance of **value-sensitive design**, stakeholder inclusion, and impact analysis.

Bias auditing and mitigation

Governance Challenges for Dual-Use NLP

- **Technical and Policy Limitations**
  - **Open Access vs. Restriction**:
    - Who gets access to powerful models (e.g., GPT-4, Claude)?
    - Trade-off between democratization and misuse control.
  - **Lack of Regulation:**
    - No binding international norms for AI safety and dual-use prevention.
  - **Responsible Publication:**
    - "Staged releases," API-gating, and content filters—often ad hoc.
- **Ethical Questions:**
  - Should researchers anticipate every possible misuse?
  - Is withholding a model ever justifiable?

Mitigation Approaches

- **Current and Emerging Solutions**
  - **Use Restrictions & Access Controls**
    - License terms prohibiting abuse (e.g., OpenAI, Anthropic, Cohere).
  - **Model Watermarking & Attribution**
    - Detectable signatures in generated content (e.g., OpenAI's watermarking research).
  - **Red Teaming and Abuse Testing**
    - Proactive evaluation by adversarial test users.
  - **Fine-tuning with Norms and Safety Filters**
    - Reinforcement learning from human feedback (RLHF) to align outputs.

Unveiling Biases in Black-Box NLP Models

- **Why It Matters:**
  - Many high-performing NLP models (e.g., GPT, BERT) are opaque: decisions are made, but rationale is hidden.
  - These models can encode and amplify societal biases across gender, race, religion, and more.
  - Uncovering bias is critical for trust, fairness, and safe deployment.
- **Challenges in Black-Box Auditing:**
  - No access to model weights or training data (API-based access only)
  - Outputs are context-sensitive and stochastic, complicating reproducibility
  - Bias may only surface in certain prompts or adversarial settings

Approaches to Auditing and Unveiling Bias

- **Input-Output Probing:** Craft controlled prompts (e.g., "The doctor said ___" vs. "The nurse said ___") and compare completions to detect preference patterns
- **Counterfactual Testing:** Modify protected attributes (e.g., gender, race, religion) in input text and observe changes in model output
- **Bias Benchmarks:** Evaluate model behavior using curated datasets:
  - *StereoSet*: evaluates stereotypical bias in language modeling
  - *CrowS-Pairs*: contrastive sentence pairs for bias evaluation
  - *BBQ*: Bias Benchmark for Question Answering
- **Influence Functions and Gradient-Based Analysis** *(for semi-black-box or white-box models):* Estimate how individual training examples affect a model's predictions
- **Representation Bias Analysis:** Investigate latent space (e.g., word or sentence embeddings) for clustering or separation across demographic groups

Governance and responsible deployment

Bias Protection in NLP

1. Data-Level Protections

- **Curated Diversity:** Ensure demographic, linguistic, and topical diversity in data sources
  - Balance across age, gender, race, geography, and socioeconomic status
- **Controlled Sampling:** Avoid over-sampling high-frequency or sensational content
  - Actively seek underrepresented voices and perspectives
- **Metadata and Transparency**
  - Use data documentation (e.g., *datasheets for datasets*) to capture source, sampling, and limitations

2. Socio-Cultural Representation

- **Context-Aware Design:** Account for culture-specific norms in politeness, discourse, and emotion
  - Avoid imposing majority-language norms as universal standards
- **Dialect and Register Inclusion:** Represent linguistic variation: AAVE, code-switching, informal registers
  - Reduce bias in tokenization and embedding models for non-standard varieties
- **Participatory Annotation:** Include diverse annotators with lived experience in marginalized communities

Bias Protection in NLP

3. Modeling Protections

- **Debiasing Word Embeddings:** Techniques like projection removal or regularization to mitigate stereotypes (e.g., Bolukbasi et al., 2016)
- **Counterfactual Data Augmentation:** Generate parallel inputs by swapping sensitive attributes (e.g., gendered names, dialects)
- **Adversarial Training & Fairness Constraints:** Train models to be invariant to protected attributes or penalize biased predictions
- **Causal Modeling & Confounder Control:** Use structured methods to isolate bias and reduce reliance on proxies

4. Evaluation and Analysis

- **Group-Level Performance Metrics:** Report precision, recall, F1 for each demographic group (not just global accuracy)
- **Bias Auditing Benchmarks:** Use datasets like **StereoSet**, **CrowS-Pairs**, **BBQ**, **ToxiGen**, and **WinoBias**
- **Error and Impact Analysis:** Evaluate *who* is misclassified and *how* misclassification impacts individuals/groups
- **Interdisciplinary Review:** Collaborate with ethicists, sociolinguists, and affected communities to contextualize model behavior

- **Takeaway:** Bias protection is not a one-time fix — it's a **continuous, multi-faceted process**. Ethical NLP requires **designing with equity in mind**, not just auditing after deployment.

Concluding Remarks

- Fixing bias is about intentional choices, continuous auditing, meaningful accountability.
- Building truly responsible NLP means transparency and preserving the social context at every step.

- Do not forget to submit your SPI

Takeaways

- NLP systems inherit and can amplify **human cognitive and social biases**.
- Misclassification and translation errors can cause **real-world harm**.
- Dual-use LLMs lower barriers for both innovation and misuse.
- Bias mitigation requires **data, modeling, and evaluation-level protections**.
- Ethical NLP demands **continuous auditing, transparency, and accountability**.